Blessed G/Michael Catholic Secondary School

First Semester Chemistry 2nd mid exam for Grade 11th

Dec, 2018 E.C Time Allowed: 1; 10'.

Name: Alukamio Menale Sec: 11 C R.No. 33

I. choose the best answer among the given alternatives

C 1. Which of the following compound does not contain an ionic bond?

A. K_2S B. NaOH C. HCl D. LiH

A 2. Which of the following ionic compound has the greatest lattice energy?

A. LiF B. LiCl D. LiBr D. LiI

D 3. Which one of the following compound does not follow the octet rule?

A. CS_2 B. PBr_3 C. IBr D. BrF_5

B 4. Which of the following molecule has a dipole moment?

A. CS_2 B. IF_3 C. BF_3 D. SF_5^+

B 5. Of the following the most polar bond is:

A. P-Cl B. Si-F C. S-O D. C-N

C 6. Which of the following molecule doesn't have a coordinate covalent bond?

A. CO B. H_3O^+ C. CO_2 D. NH_4^+

B 7. How many bonding pairs and lone pairs, respectively does the ion XeF_4?

A. 3,2 B. 4,2 C. 5,1 D. 4,1

C 8. Which of the following molecule represents a nonpolar covalent bond?

A. B-Cl B. C-Cl C. Cl-Cl D. Mg-Cl

D 9. Which of the following is not a form of chemical bonding?

A. covalent bonding B. Metallic bonding C. Ionic bonding D. Hydrogen bonding

A 10. Which of the following is not a property of an ionic compound?

A. malleable B. Brittle C. hard D. High melting point

D 11. Which substance does not conduct electricity?

A. Na(s) B. molten $pbBr_2$ C. NaCl(aq) D. $CaCl_2$(s)

A 12. When a student draws a plausible Lewis structure for hydrazine molecule (N_2H_4), how many lone pair of electrons are available?

A. 2 B. 1 C. 3 D. 4

C 13. Which of the following molecule represents a nonpolar covalent bond?

A. B-Cl B. C-Cl C. Cl-Cl D. Mg-Cl

B 14. How many bonding pair and lone pair on a central atom, respectively does the ion ICl_4^- have?

A. 2, 2 B. 4, 2 C. 5, 1 D. 4,1

D 15. Which of the following ionic compound does not obey the octet rule?

A. MgO B. KF C. $CaCl_2$ D. ZnF_2

A 16. Which of the following compound would be expected to have the lowest melting point?

A. BaI_2 B. BaF_2 C. $BaBr_2$ D. $BaCl_2$

A 17. Which of the following molecule or ion will not exhibit delocalized bonding?

A. NH_4^+ B. NO_3^- C. N_3^- D. CO_3^{2-}

A 11. Which of the following elements has the lowest first ionization energy?
A. Potassium B. Sodium C. calcium D. Argon

B 12. A neutral atom "x" at its ground state has a half filled 5p orbital. What is the atomic number of x?
A. 63 B. 51 C. 54 D. 60

B 13. Which of the following is permissible set of quantum number?
A. $n=3, \ell = 2, m\ell = 3, m_s = +\frac{1}{2}$
B. $n=2, \ell = 0, m_\ell = 0, m_s = +\frac{1}{2}$
C. $n=0, \ell = 0, m\ell = 0, m_s = +\frac{1}{2}$
D. $n=4, \ell = 3, m\ell = -3, m_s = +\frac{1}{2}$

D 14. Which pair of atomic orbitals possesses the same number of radial nodes?
A. 3s and 3p B. 4s and 3s C. 3s and 3d D. 1s and 3d

C 15. How many orbitals are in an atomic sublevel with $\ell = 3$?
A. 3 B. 5 C. 7 D. 9

D 16. According to Hund's rule degenerate energy orbitals are filled with;
A. Two electrons of parallel spins placed in the first orbital
B. Maximum number of unpaired electrons with opposite spins
C. Two electrons of opposite spins placed in the first orbitals
D. Maximum number of unpaired electrons with parallel spin.

C 17. The possible value of ℓ in 4f sublevel is/are;
A. 0,1,2,3 B. 1,2,3 C. 3 D. 0,1,2

A 18. _____ orbitals are spherically symmetrical.
A. s B. p C. d D. f

B 19. All orbitals can hold a maximum of _____ electrons.
A. 1 B. 2 C. 3 D. 4

A 21. The possible quantum number values for an orbital in the 5f sub shell will be.
A. n =5, l =3, ml= -3, -2, -1, 0, +1, +2, +3
B. n =5, l=2, ml = -2,-1, +1, +2
C. n =5, l = 2, ml =-2, -1, 0, +1, +2
D. n =5, l =3, ms = -3, -2, -1, 0,+1, +2, +3

B 22. Which one of the following would have the smallest ionic radius?
A. $_8O^{2-}$ B. $_9F^-$ C. $_7N^{3-}$ D. $_{17}Cl^-$

D 23. The maximum number of electron in n=5, $m\ell = 0$ is;
A. 2 B. 6 C. 8 D.10 E.50

E 24. How many orbitals are associated with the 5f sublevel?
A. 25 B. 16 C. 3 D. 5 E.7

A 25. Which of the following species would require the highest energy for the removal of electron?
A. Mg^{2+} B. Na^+ C. Ne D. F

Blessed G/Michael Catholic Secondary & Higher Education Preparatory School

First Semester Chemistry test for Grade 11th

Jan, 2018 E.C

Time Allowed:1hr.

Name: Mulugeta Menale Sec: 11C R.No. 33

Choose the best answer among the given alternatives

D 1. How many bonding pairs and lone pairs, respectively does the ion XeF_4?
A. 3,2 B. 4,2 C. 5,1 D. 4,1

A 2. Which of the following molecule does not have a tetrahedral central atom?
A. SF_4 B. AlH_4^- C. BF_4^- D. $SiCl_4$

D 3. Which of the following is not a form of chemical bonding?
A. covalent bonding B. Metallic bonding C. Ionic bonding D. Hydrogen bonding

C 4. Which of the following has the largest dipole moment?
A. CO B. HCN C. HF D. HCl

A 5. Which of the following molecule has a dipole moment?
A. SO_2 B. CH_4 C. BF_3 D. CO_2

B 6. Which of the following molecule has a dipole moment?
A. XeF_2 B. IF_3 C. BF_3 D. SF_5^+

C 7. What kind of hybrid orbitals are utilized by the carbon atom in CF_4 molecule?
A. SP B. SP^2 C. SP^3 D. SP^3d

B 8. There is a progressive decrease in the bond angle in the serious of molecules CCl_4, PCl_3, and H_2O. According to the VSEPR model, this is best explained by:-
A. Increasing electronegativity of the central atom.
B. Increasing number of lone pair electrons.
C. Decreasing size of the central atom.
D. Decreasing bond strength

C 9. What is the molecular shape of ICl_4^-
A. Octahedral B. T- shaped C. square planar D. Trigonal bipyramid

A 10. Which of the following statements is correct about nitrosyl chloride /NOCl/?
A. It has a bent geometry with N a central atom
B. It has a bent geometry with O a central atom
C. It has a trigonal planar geometry with N a central atom.
D. It has a trigonal planar geometry with O a central atom.

D 11. Which one of the following compound does not follow the octet rule?
A. CS_2 B. PBr_3 C. IBr D. BrF_5

B 12. Which of the following molecule has a dipole moment?
A. XeF_2 B. IF_3 C. BF_3 D. SF_5^+

D 13. Which of the following molecule is planar?
A. SF_4 B. NF_3 C. SiF_4 D. XeF_4

B 14. Which is the T- shaped molecule;
A. ICl_4^- B. ClF_3 C. $CHCl_3$ D. PCl_5

25

25

Excellent

15. Which one of the following molecule is trigonal bipyramid?

A. CH_4 B. BF_3 C. SF_4 D. PCl_5

16. Predict which one is non- linear molecule

A. HCN B. CO_2 C. H_2O D. C_2H_2

17. What is the shape of the sulphate ion?

A. Tetrahedral B. square planar C. hexagonal D. Trigonal bipyramid

18. Which of the following molecules that includes a central atom surrounded by four peripheral atoms Would be considered as polar?

1. SF_4/see saw/ 2.CH_2Cl_2/tetrahedral/ 3.XeF_4 /square planar/

A. 1 only B. 2 only C. 3 only D 1 and 2 E. All

19. Which of the following molecules would you expect to be non- planar?

A.CCl_4 B. BCl_3 C. XeF_4 D. SO_3

20. Which of the following molecule doesn't have a coordinate covalent bond?

A. CO B. H_3O^+ C. CO_2 D.NH_4^+

21. If a molecule has a trigonal pyramidal shape how many non- bonding pair of electron are present in the valance shell of the centeral atom?

A. 1 B. 2 C. 3 D. 4

22.Selenium (Se) is a group "VIA" element. What will be the molecular geometry and number of lone pair electrons respectively that exists in the ion $[SeCl_4]^{2-}$?

A. See saw, 1 B. square planar, 2 C. see saw, 2 D. linear, 3

23. How many bonding pairs and lone pairs, respectively does the ion XeF_4?

A. 3,2 B. 4,2 C. 5,1 D. 4,1

24. Which of the following molecule does not have a tetrahedral central atom?

A. SF_4 B. AlH_4^- C. BF_4^- D. $SiCl_4$

25. Which of the following is not a form of chemical bonding?

A. covalent bonding B. Metallic bonding C. Ionic bonding D. Hydrogen bonding